



सत्यमेव जयते

NITI Aayog

DRAFT



SECTOR AT A GLANCE

Commercial Crops

Agriculture and Allied Sectors in India

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Agriculture and Allied Sectors in India

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Commercial crops are widely harvested due to their higher economic returns and use in major manufacturing sectors such as food processing and textiles. Cotton, sugarcane and jute, identified as priority crops by the National Food Security Mission¹, along with copra/coconut and tobacco are the major commercial crops¹ integral to the government's strategic efforts to enhance farmers' incomes, retain soil fertility and drive better land and water-use efficiency. Commercial crops account for about 9% of the agricultural economy in India, with sugarcane and cotton being the highest contributors; these two crops are produced in several states and account for 94.2% of the export contribution from commercial crops.² On the other hand, jute and coconut are more concentrated in the eastern and southern regions of India respectively.

Commercial crops are input-intensive and face several production constraints such as rising labour costs, high water footprint, widespread diseases, pest attacks and low irrigation channels, and at a market-level, price volatility further affects farmers' income security. Therefore, the government has focused on providing crop-specific insurance, anchoring technology missions to develop processing infrastructure, and improving long-term economic viability through measures such as the Jute Diversification scheme. For sugarcane, which often experiences surplus production, the government has introduced the Ethanol Blended Programme to protect remuneration for farmers. The water-intensive crop is also a major focus for the 'Per Drop More Crop' component of Pradhan Mantri Krishi Sinchayee Yojana.

Figure 1: Major Commercial Crops of India²



Note: Production values stated are averages over the period of 2017-22.
*Tobacco too falls under the ambit of commercial crops in India but the scope of the document confines itself to the four major commercial crops given their overall importance in terms of both economic contribution and policy priorities.

Apart from being vital raw materials for the domestic manufacturing sector, commercial crops like jute and sugarcane have considerable trade value in international markets and account for 15-16% of India's global export value contribution.

Figure 2: Overview of the Economic Contribution and Production-levels of Commercial Crops in India



Source: Estimates based on National Accounts Statistics data, Ministry of Statistics and Programme Implementation (MoSPI), Government of India
*GVA refers to Gross Value Added
**CAGR refers to Compound Annual Growth Rate

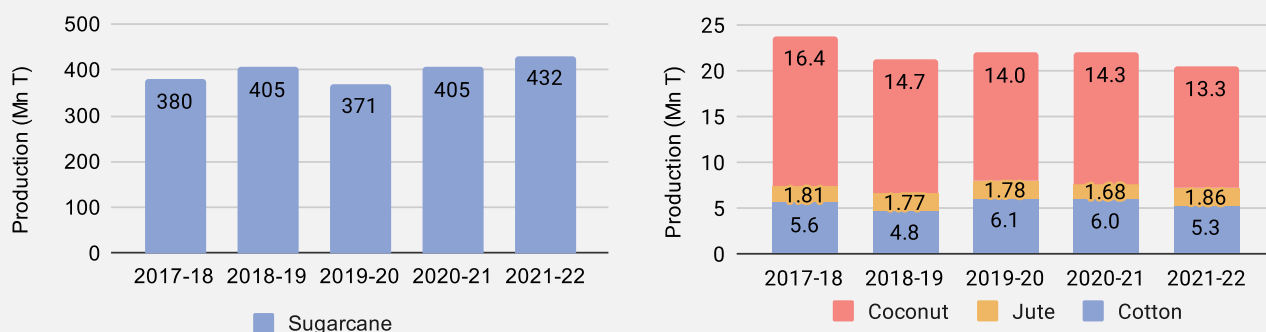
1 Ministry of Agriculture & Farmers Welfare. (2018-19). *National Food Security Mission —Operational Guidelines*.

2 Ministry of Agriculture & Farmers Welfare. (2022). *Agricultural Statistics at a Glance*. (This document has been referred to as the primary source for all the crop-specific data on production, area, yield and trade. The 2021-22 figures mentioned throughout the document are based on fourth advance estimates declared by the Ministry of Agriculture on 11.08.2021. In addition, the 2021-22 figures for both exports and imports are projected values.)

Production Trends

The production levels of commercial crops have witnessed a stagnation in the past five years (Figure 3). In 2021-22, the total output of the four commercial crops was 452 million tonnes, with sugarcane contributing to 95% of this value; sugarcane also experienced the largest growth in cultivation area.² With the exception of coconut, the yield has increased for all commercial crops with jute recording the highest CAGR of 2%. Uttar Pradesh (39.3%), Gujarat (4.2%), Tamil Nadu (4.2%), Bihar (2.9%), and Andhra Pradesh (1.1%), are some of the states that are the top producers of commercial crops.²

Figure 3: Production Trends for Major Commercial Crops (2017-18 to 2021-22)



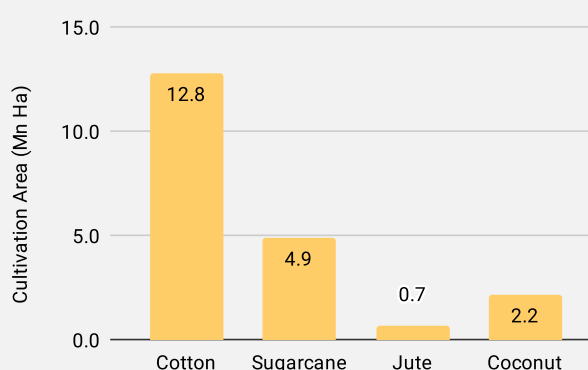
Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

The total estimated area used for the cultivation of commercial crops is 20 million hectares.²

Sugarcane demonstrated the highest growth in production among all the commercial crops with a CAGR of 2.6%.

Cotton, which utilises the highest proportion of cultivated land, has seen a decline in its gross area coverage since 2017.

Figure 4: Average Cultivation Area (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Figure 5: Yield Comparisons — India vs. Global

	India (2021) ²	Global Average (2021)	Country with the Highest Yield
Cotton	445 kg / ha	775 kg / ha	China
Sugarcane	83,887 kg / ha	70,570 kg / ha	Peru
Jute	2,709 kg / ha	2,246 kg / ha	China
Coconut	8,966 nuts / ha	5,440 kg / ha	Jamaica

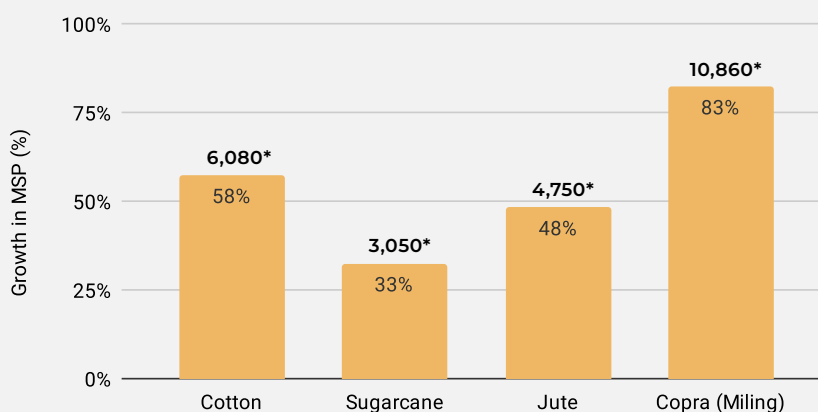
Source: Food and Agriculture Organization; Directorate of Jute Development, Department of Agriculture & Cooperation; Coconut Development Board

Consumer preferences and price signals have proven to be major determinants of commercial crop production. Jute and copra have witnessed the highest surge in market prices as depicted by their inflation rates since 2011 (Figure 7). These crops face multiple supply-side constraints and rely on imports to fulfill domestic demand.

The average per capita availability of commercial commodities like sugar and cotton has remained stable in the past few years (19.9 kg/year and 25.8 kg/year in 2021-22 respectively).² In the case of jute and copra, the domestic demand has grown and the production levels have not been able to meet market needs. This supply-demand gap has reduced the per capita availability of these crops and led to an increase in market prices.

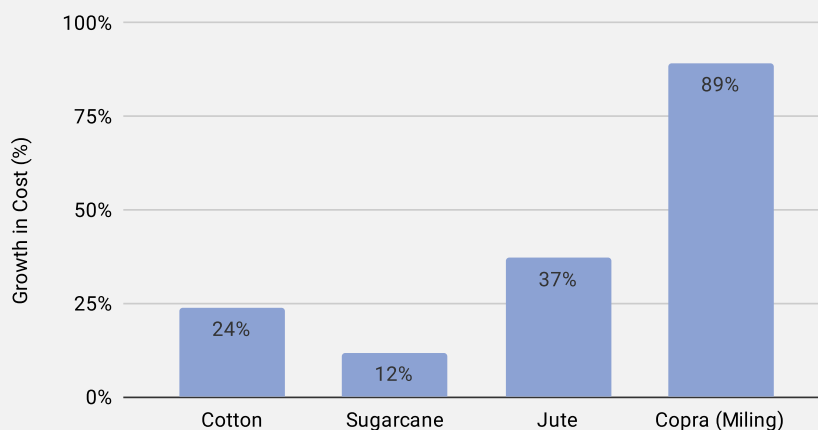
Price support by the government has ensured a minimum margin of 50% over the rising production cost for farmers (Figure 6). The high expenditure of cultivating commercial crops can be attributed to rising labour costs, limited access to quality seeds, extensive use of water and fertilisers, and prevalent disease and pest attacks during the growth period. This also hinders India's potential to be at par with average global productivity despite its status as one of the largest producers of commercial crops.

Figure 6: Change in MSP* over Time (2016-17 to 2021-22)



Source: Cabinet Committee on Economic Affairs (CCEA)
*Minimum Support Price (2022-23)

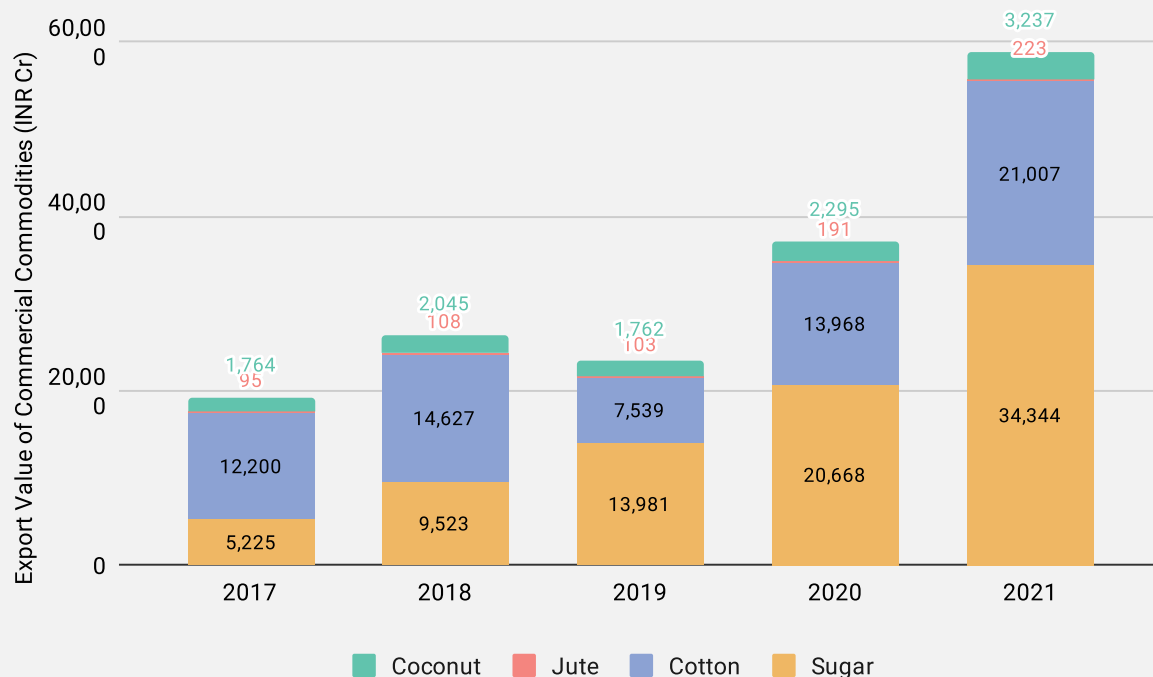
Figure 7: Change in Projected Production Cost (2016-17 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Since 2017, commercial crops have witnessed a steady growth of 20% in terms of export value (INR). Sugar exports accounted for approximately 60% of the total trade in commercial crops.² Cotton too has been a significant contributor to global exports. In 2020, India was the third largest exporter of raw cotton globally, accounting for about 10.2% of the total global exports.

Figure 8: Export Value of Commercial Commodities (2017-18 to 2021-22)



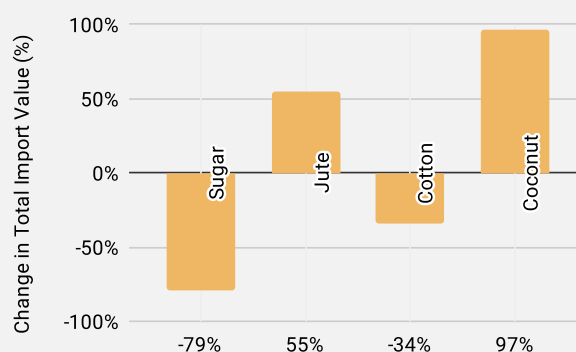
Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Despite being the largest producer of coconut and jute in the world, India still witnessed an overall rise in imports for these crops. The import of cotton, on the other hand, has been on a decline since 2017. While sugar imports experienced a rise during the year 2020-21 by substantial margins.³

Jute imports have risen by over 50% over the past five years. However, this increase has been volatile, with a spike in 2019, followed by a decline until 2021.

Coconut imports have nearly doubled over the past five years. However, imports dropped by 37% in 2021-22.

Figure 9: % Change in Total Import Value (2017-21)



Source: Ministry of Commerce & Industry, Government of India

³ V Kumar. (2021). Trends and Performance of India's Agricultural Trade in the Midst of COVID-19 Pandemic

Cotton

India accounts for 25% of the global cotton production, strongly contributing to India's foreign exchange through exports of raw cotton, intermediate products like yarn and fabrics, and finished garments.¹ Despite utilising the largest cultivation area among all the commercial crops, cotton production has not shown a significant increase, with the total output being approximately 5.6 million tonnes in 2021-22. Around 67% of India's cotton is grown in rainfed areas and 33% in irrigated areas. In terms of productivity, India ranks 38th globally with a yield of 445 kg/ha.²

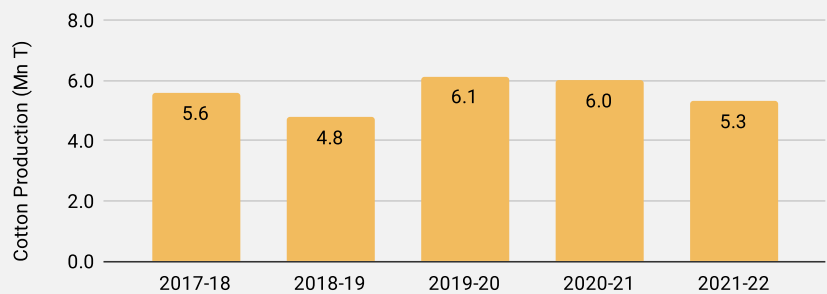
5.3 Mn T²

Total Production (2021-22)

-1.24%

CAGR (2017-18 to 2021-22)

Figure 10: Cotton Production in India (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022),
Ministry of Agriculture and Farmers' Welfare, Government of India

Cotton is mainly grown in three agro-ecological zones in India:^{4 5}

Northern Zone:

Punjab, Haryana and Rajasthan

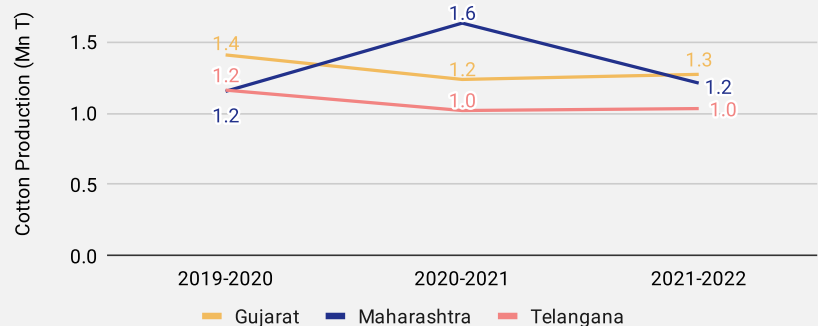
Central Zone:

Gujarat, Maharashtra and Madhya Pradesh

Southern Zone:

Telangana, Andhra Pradesh, Karnataka and Tamil Nadu

Figure 11: Top 3 Cotton-Producing Indian States (2019-20 to 2021-22)



Source: Agricultural Statistics at a Glance (2022),
Ministry of Agriculture and Farmers' Welfare, Government of India

Cotton cultivation has seen some significant deviations in top producing states like Maharashtra:

- In 2020-21, Maharashtra was the largest cotton producer (1.6 million tonnes), but Gujarat surpassed it in 2021-22, producing 1.3 million tonnes.
- In 2021-22, Maharashtra registered a considerable fall in production of about 0.6 million tonnes. One of the significant contributing factors for this decline was the persistence of frequent infestations by pests like the pink bollworm.

⁴ Ministry of Textiles (2022). *Cotton Sector*. Retrieved 4 October, 2023.

⁵ As per the third advance estimates released by the Ministry of Agriculture and Farmers' Welfare in May 2023, the total cotton production was 34.3 million bales (of 170 kg each). This amounts to approximately 5.8 million tonnes.

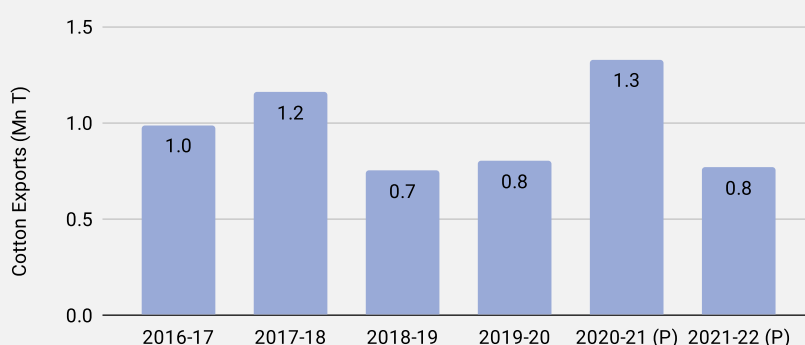
Measures taken to strengthen the cotton sector:

- **Consistent market support:** The government has ensured the economic viability of cotton by establishing an MSP and enabling direct procurement through the Cotton Corporation of India. In 2023-24, the MSP for medium staple cotton and long staple cotton were raised to INR 6,620 and INR 7,020 respectively.
- **Mainstreaming of Bt Cotton:** Bt Cotton, a genetically-modified variety resistant to pest attacks, was approved for commercial use in 2002 which led to a doubling of cotton output between 2002 and 2019. The launch of the Technology Mission on Cotton fostered the development of high-yielding varieties, better farm management practices and increased area of cultivation for Bt cotton yields.
- **Intercropping:** Through Front Line Demonstrations (FLDs), the government is encouraging the intercropping of cotton with other crops such as pulses, cereals and oilseeds to improve economic returns for farmers.

India's Cotton Exports and Imports

India is **the third largest exporter of cotton, followed by the United States of America and Brazil, exporting an estimated 45 lakh bales (0.76 million tonnes) in 2021-22** equivalent to **8% of the global export amount** of 597 lakh bales (10.15 million tonnes).⁶

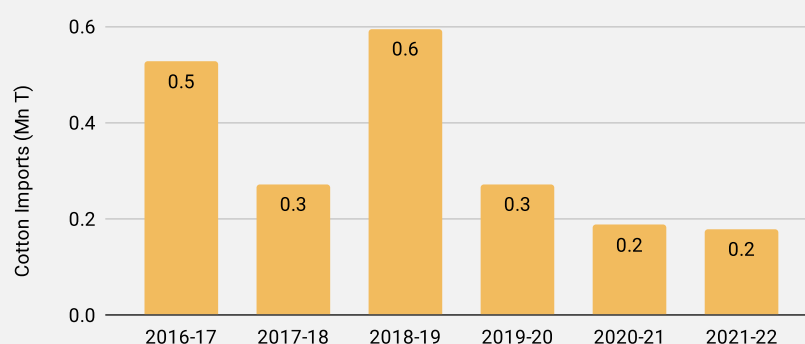
Figure 12: India's Export of Cotton (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

India's cotton imports have been on a **decline** since **2018**, with less than **10% of the country's total consumption** imported by the textile industry to meet the growing domestic demand.

Figure 13: India's Import of Cotton (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

6 Statista. (2023). *Leading cotton exporting countries*. Retrieved July 2023.

Sugarcane

In 2022, the global sugar production amounted to approximately 177 million tonnes. India is the second largest producer of sugar in the world after Brazil and is also the largest consumer. The Indian sugar industry's annual output is worth approximately INR 80,000 crores.⁷ Sugarcane impacts the livelihoods of nearly 50 million farmers while also generating an additional 0.5 million jobs across the domestic value chain. It also enables employment in ancillary activities such as transport, trade servicing of machinery, and supply of agriculture inputs. In 2021-22, around 432 million tonnes of sugarcane were produced, generating about 39.4 million tonnes of sugar (sucrose).⁷

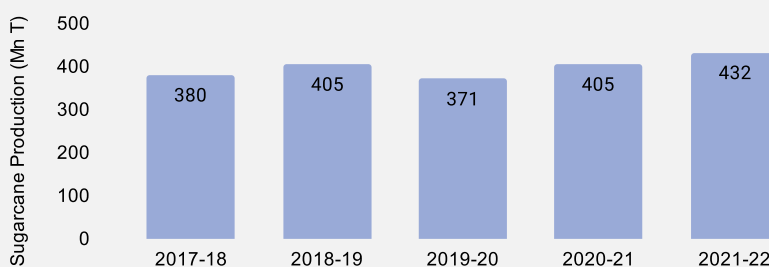
432 Mn T⁸

Total Production (2021-22)

2.5%

CAGR (2017-18 to 2021-22)

Figure 14: Sugarcane Production in India (2017-18 to 2021-22)

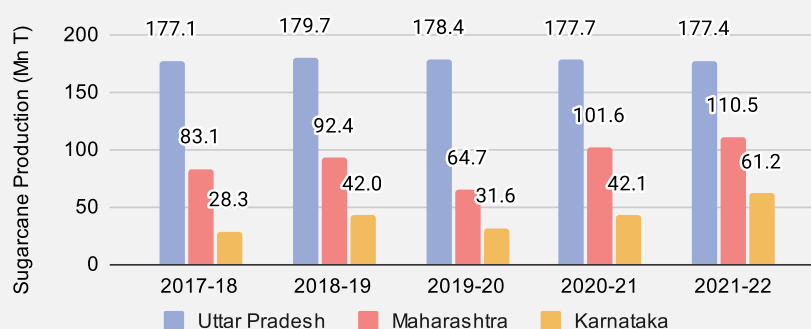


Source: Agricultural Statistics at a Glance (2022),
Ministry of Agriculture and Farmers' Welfare, Government of India

Uttar Pradesh remains the largest sugarcane producer in the country, producing ~177.4 million tonnes of the crop in 2021-22.

Both **Maharashtra** and **Karnataka** have shown a growth in sugarcane production with a CAGR of 5.9% and 16.7% respectively.

Figure 15: Top 3 Sugarcane-Producing Indian States (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022),
Ministry of Agriculture and Farmers' Welfare, Government of India

Measures taken to strengthen the sugarcane sector:

- **Sugar and Sugarcane Development Fund:** Due to persistent uncertainty in sugar prices and rain-fall during the harvest season, the liquidity of sugar mills has remained a major cause of concern, prompting the government to create various liquidity support measures.
- **Replacement of Statutory Minimum Price (SMP) with Fair and Remunerative Price (FRP):** Under the FRP, farmers are not required to wait till the end of the season or for profit announcements by sugar mills or the government. The new system assures margins, irrespective of whether the mills generate the required profit.
- **Introduction of high-yielding varieties:** The growing adoption of cultivars with higher sugar recovery rates along with technological advances have helped improve sugarcane productivity.
- **Minimum distance criterion:** Through the Sugarcane Control Order, the central government has prescribed a minimum radial distance of 15 km between any two sugar mills, potentially ensuring a minimum availability of cane for all mills.⁹

⁷ Ministry of Consumer Affairs, Food & Public Distribution. (2023). *Sugar and Sugarcane Policy*. Retrieved 13 September 2023.

⁸ As per the third advance estimates released by the Ministry of Agriculture & Farmers' Welfare in May 2023, the total production of sugarcane was 494.2 million tonnes, which was 14% higher compared to the previous year.

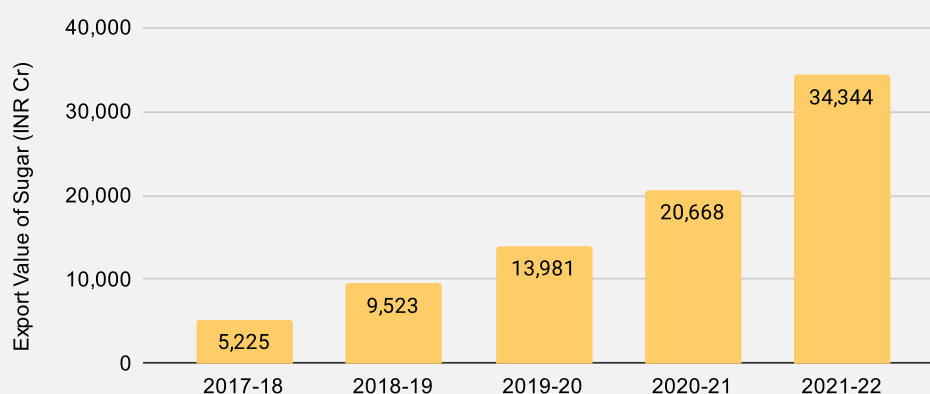
⁹ PRS Legislative Research. *Regulation of Sugar Sector in India*. (2012). Retrieved July 2023.

India's Sugarcane Exports and Imports

As per the data from Ministry of Commerce & Industry, India has consistently been one of the major exporters of sugar globally (second after Brazil) due to consistent surplus production, exporting predominantly to Indonesia, Bangladesh, Sudan and the United Arab Emirates.

Between 2017-18 and 2021-22, India's sugar exports grew at a CAGR of 45.7% (Figure 16).

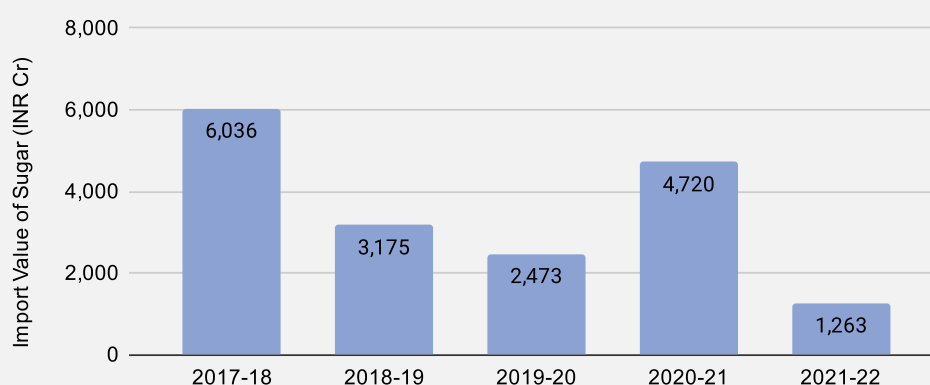
Figure 16: India's Export of Sugar (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

The previously fluctuating volume and value of sugar imports has recently experienced a considerable decline due to growing domestic production. In 2021-22, sugar imports in India fell by 73%² (Figure 17).

Figure 17: India's Import of Sugar (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Jute and Mesta

In commercial use, the crops jute and mesta are together known as raw jute with nearly identical applications. Like cotton, jute is one of the most versatile and eco-friendly raw materials having a total cultivation area of 0.69 million hectares as of 2021-22. The jute industry in India produces a diverse range of products and employs a workforce of approximately 3.5 lakh people.¹⁰ **West Bengal (76.5%), Assam (8.3%), and Bihar (7.5%) are the major producers of jute in India.** Mesta, on the other hand, is grown all over the country. According to the Ministry of Agriculture and Farmers' Welfare, jute is mostly a rainfed crop with only 15% of its cultivation area requiring irrigation. In 2008, an Act was passed by the Government of India to establish National Jute Board who acts as the central body to implement programmes and disseminate new technologies pertaining to the development of jute economy in India.

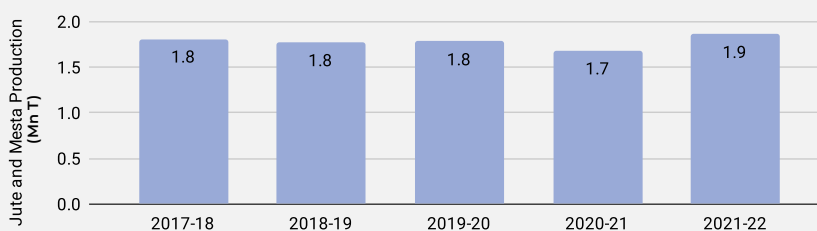
1.9 Mn T¹¹

Total Production (2021-22)

1.1%

CAGR (2017-18 to 2021-22)

Figure 18: Jute and Mesta Production in India (2017-18 to 2021-22)

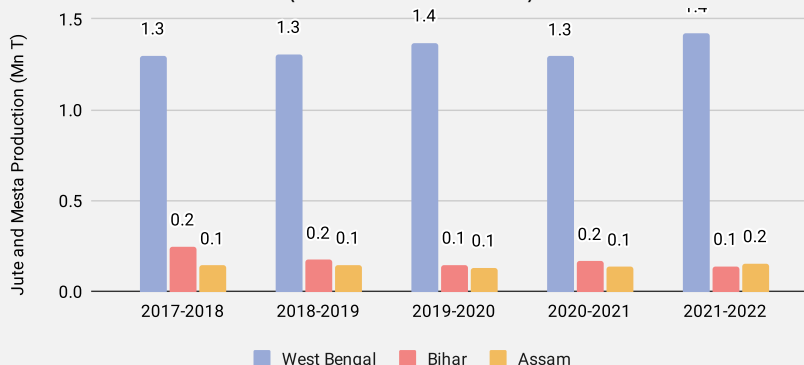


Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

The Indian jute industry is highly concentrated in West Bengal due to its prevalence of mills. However, despite being the top producer, the state has not experienced any significant growth in its total annual jute output.

In 2020-21, West Bengal, Assam and Bihar accounted for 98% of India's total jute production. Other notable jute producers include Tripura, Jharkhand, Orissa, Nagaland and Meghalaya.

Figure 19: Top 3 Jute- and Mesta-Producing Indian States (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Measures taken to strengthen the jute sector:

- **MSP:** Whenever the market price of raw jute falls below a certain level, the Jute Corporation of India (JCI) procures raw jute at Minimum Support Price (MSP), fixed based on the recommendations of the Commission for Agricultural Cost and Prices (CACP).
- **Modernisation:** Technological upgradation in jute mills through schemes like the Incentive Scheme for Acquisition of Plants and Machinery (ISAPM) has led to the manufacturing of value-added biodegradable Jute Diversified Products at scale.
- **Jute: Improved Cultivation and Advanced Retting Exercise (Jute-ICARE):** The Jute-ICARE scheme provides multiple benefits to jute producers such as certified seeds at subsidised rates, seed drills to facilitate line sowing, nail-weeders to carry out periodic weeding, and retting technologies under water scarce situations.¹²

¹⁰ India Brand Equity Foundation. (2023). *Jute Industry and Exports*. Retrieved on July 2023.

¹¹ As per the third advance estimates released by the Ministry of Agriculture and Farmers' Welfare in May 2023, the total jute production was 1.6 million tonnes, which was 11% lower compared to the previous year.

¹² Ministry of Textiles. (2017). *Schemes and Measures to Strengthen Jute Sector*. Retrieved July 2023

India's Jute Exports and Imports

Globally, India remains the leading exporter of jute with a total export value of INR 223 Cr in 2021-22² (Figure 20). Between 2015-16 and 2021-22, the country's jute exports grew at a CAGR of 10%. Fibre processing is a key definer of India's export success as several high-quality products, such as carpets, decorative fabrics and handbags, are traded in international markets. As of 2021-22, 58% of the country's jute is exported to the United States of America (Figure 21).

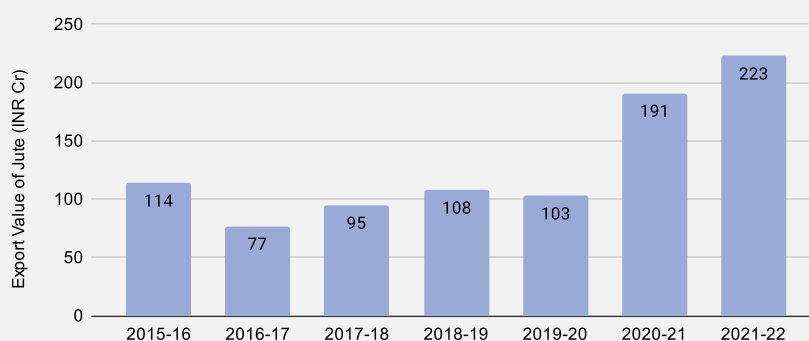
The United States of America was the leading importer of India's jute products in 2021-22. The USA, along with the UK and Germany, constitutes 88% of India's total jute export value.

Egypt, Germany, Italy, Japan, Saudi Arabia, and Turkey are some of the other major export destinations for India's jute and jute products.

India imports both raw jute and jute products, especially from the neighbouring countries of **Bangladesh, Nepal and China**.

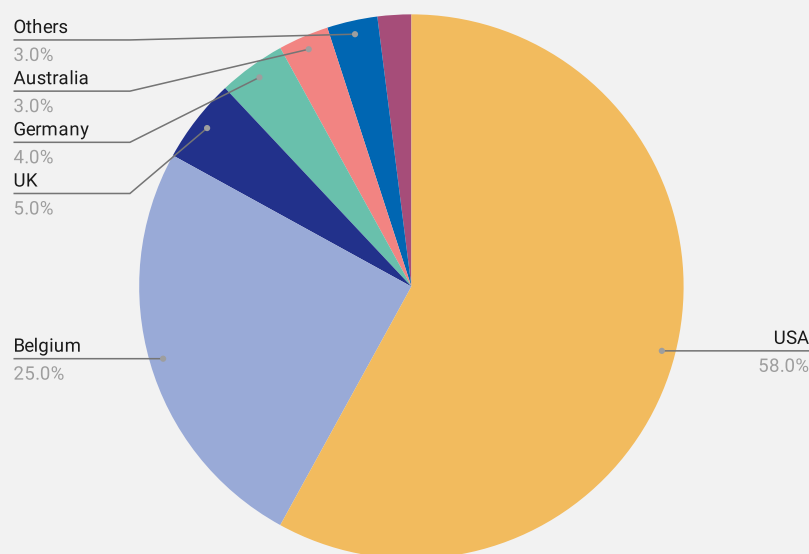
In 2021-22, India imported about **62.5 thousand tonnes** of raw jute². Additionally, the total import value of hessian, one of the major jute products, was **INR 363.4 Cr** in the same year².

Figure 20: India's Export of Jute (2017-18 to 2021-22)



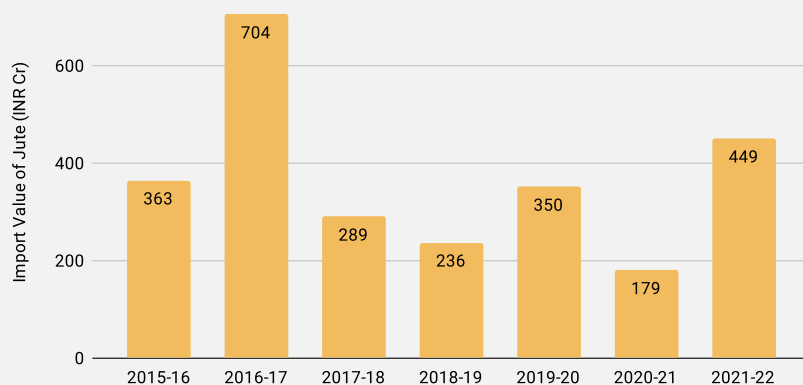
Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Figure 21: India's Top Jute Export Destinations



Source: National Jute Board

Figure 22: India's Import of Jute (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Coconut*

India is the largest coconut-producing country in the world, accounting for about **31.4% (13.3 million tonnes)** of the world's total production during 2021-22¹³ (Figure 23). During **2022-23**, the crop contributed ~INR **30,796 crores** to the nation's Gross Domestic Product (GDP)¹³. Compared to average global productivity, India's yield is higher than most countries at 8,966 nuts/ha but has experienced a slight decline since 2016-17. Some of the traditional coconut-based activities in the country include **copra-processing, coconut oil extraction and coir** manufacturing. Coconut production is largely concentrated in the southern states with Kerala, Karnataka and Tamil Nadu accounting for 81% of the country's coconut production in 2021-22.¹³ In 1981, the Government of India established Coconut Development Board (CDB) as an apex statutory body to enable value chain development, product diversification and technological innovation within the coconut economy. The board operates under Ministry of Agriculture and Farmers Welfare and looks after the entire ambit of activities associated with coconut production, processing and exports.

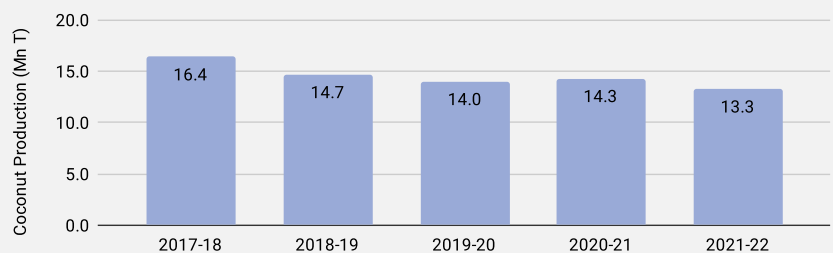
13.3 Mn T²

Total Production (2021-22)

-4.1%

CAGR (2017-18 to 2021-22)

Figure 23: Coconut Production in India (2017-18 to 2021-22)

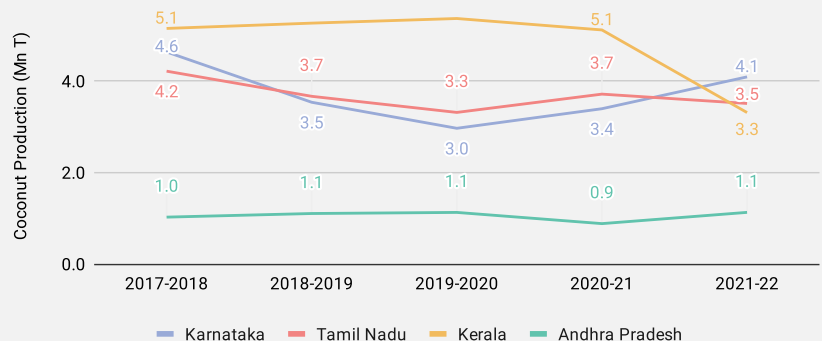


Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

The southern states of **Karnataka (30.8%), Tamil Nadu (26.4%), Kerala (24.9%) and Andhra Pradesh (8.5%)** cumulatively account for 90.6% of total coconut production.

Karnataka surpassed Kerala in 2022 to become the leading producer of coconut in the country (Figure 24).

Figure 24: Top 4 Coconut-Producing Indian States (2017-18 to 2021-22)



Source: Agricultural Statistics at a Glance (2022), Ministry of Agriculture and Farmers' Welfare, Government of India

Measures taken to strengthen the coconut sector:

- **Area expansion and productivity enhancement:** The CDB runs multiple financial and non-financial assistance schemes for coconut production and adoption of better coconut-farming technologies.
- **Supply chain development:** Given the demand for a variety of coconut products, the government has focused on augmenting processing facilities across the country. For example, Technology Mission on Coconut provides financial assistance of 25% for setting up coconut processing units with a maximum budget of Rs 50 lakh.
- **Disease control, diversification and market support:** The government has promoted technologies for **managing insect pests and disease-affected coconut gardens**, processing and product diversification, and market research and promotion.¹⁴

¹³ India Brand Equity Foundation. (2023). *Coconut Industry and Exports*. Retrieved on July 2023.

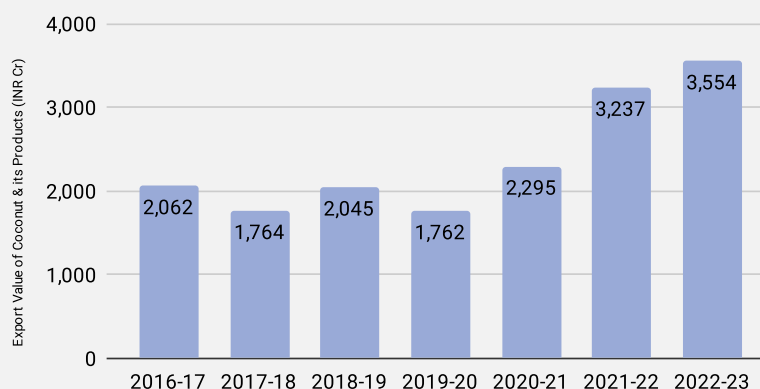
¹⁴ Coconut Development Board. (2021). *Climate change, Carbon sequestration and Coconut based Ecosystem*. Retrieved on July 2023.

*Even though the coconut is categorised as a plantation crop under horticulture, this document has included it as it is an important source of income and profits for farmers as well as a raw material for copra (a major commercial crop in India as per CACP).

India's Coconut Exports and Imports

India is the largest coconut-producing country in the world, exporting products worth approximately INR 3,236 crores in 2021-22¹⁵ (41% increase from the previous year). It exports a variety of coconut-based products to more than 140 countries with major export destinations being the United Arab Emirates, Saudi Arabia, the United States of America, and Vietnam.¹⁵ Previously an importer of coconut oil from Asian countries like Sri Lanka and Malaysia, India has grown to become an exporter of the product to these countries since 2016. Although coconut exports experienced fluctuations between 2016-17 and 2020-21, largely due to a decline in domestic production levels and crop productivity, exports grew by 24% from 2020-21 to 2022-23¹⁵ (Figure 25).

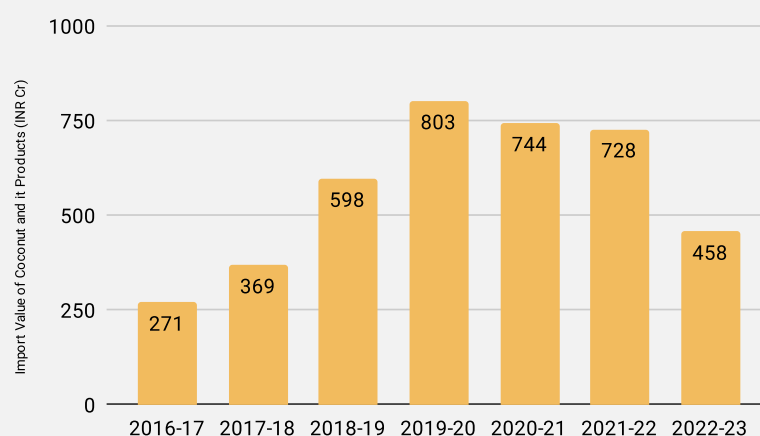
Figure 25: India's Export of Coconut and Coconut Products (2016-17 to 2022-23)



Source: Coconut Development Board

India imports coconut from 34 nations in all, with **Bangladesh, Indonesia, Philippines, Sri Lanka and China** accounting for up to **98.55%** of the total imports. Between 2016-17 and 2019-20, India's import dependency grew by a CAGR of 43.6% due to a 5-6% fall in domestic production. Between 2020-21 and 2022-23, however, imports have declined by **21.6%**.¹⁵ (Figure 26).

Figure 26: India's Import of Coconut and Coconut Products (2016-17 to 2022-23)



Source: Coconut Development Board

¹⁵ Coconut Development Board. (2021). *Export and Import of Coconuts in India*. Retrieved July 2023

Emerging markets for commercial crops are likely to grow and encourage greater use of technology



Market expansion for organic cotton

- In recent years, the demand for organic cotton in India and other South Asian economies has surged, resulting in more investments from private brands to build future capacities of this crop.
- Organic and in-conversion cotton production in India has also expanded. According to the Ministry of Textiles, the production of organic cotton during 2020-21 has been 0.81 tonnes compared to only 0.3 tonnes during 2018-19.¹⁶
- Adapting to Voluntary Sustainability Standards (VSS) through formal certifications can help India remain globally competitive in the cotton supply chain and strengthen its position in the export market.¹⁷



Strategic pivots to utilise sugar surplus for biofuel energy¹⁸

- Since 2019, India has dealt with its sugar surplus by exporting a large portion of it. But lately, the government has applied a strong thrust for the diversion of sugarcane juice into the production of ethanol. The latest estimates show a sugar surplus of 143 million tonnes for sugarcane production.¹⁸
- India is aiming to prioritise ethanol production and expand its sugar mills through a subsidy scheme. In 2021, India used 3.6 million tonnes (about 9%) of its sugar for the manufacture of ethanol, a rising trend expected to invite more players in the processing and value-addition stages.¹⁸



Modernisation and product diversification in the jute industry

- With more consumers opting for eco-friendly alternatives, a greater demand for value-added jute products like woven and non-woven jute geotextiles, jute agro-textiles, jute-blended fibre yarn, river embankments, and lifestyle products, may be imminent.¹⁹
- Given stringent competition from other Asian countries like Vietnam and Bangladesh, there is a greater scope to invest in mills with modernised facilities, research and development, and supply chain solutions within the Indian jute industry.



Research and value chain development for coconut harvesting

- Over the years, coconut production has seen a considerable decline due to the root wilt disease, one of the most damaging conditions for the crop. Funds from the Technology Mission on Coconut have been earmarked to support the demonstration of proven disease- and pest-management technologies.
- The Coconut Development Board is also placing greater emphasis on enhancing processing units to support product diversification and growth in export markets. This growing focus is expected to invite more projects in this area and encourage public-private partnerships.

¹⁶ National Centre for Natural and Organic Farming. (2021). *Organic Farming Newsletter*. (*As per India Standards for Organic Textiles, organic cotton is categorised as a fibre to convey that the process of manufacturing has been carried out in a socially- and environmentally-sound manner)

¹⁷ Observer Research Foundation. (2022). *India's cotton production: Seeking white gold*.

¹⁸ Ministry of Consumer Affairs, Food and Public Distribution. Retrieved August 2023.

¹⁹ CareEdge. (2023). *Modernisation and Diversification in Jute Industry – Pathway to the 5F Vision*.



20 Ministry of Agriculture and Farmers' Welfare. (2022). *Doubling sugarcane farmers' income by 2022 – strategies and way forward*.
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 22 A Qureshi & J Bhakay. (2021). *Review Of Textile Trade Between India, It's Neighbours And The ASEAN Countries*.
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 25 Ministry of Agriculture and Farmers' Welfare. (2017). *Price Support Scheme (PSS): The Operational Guidelines*.
 26 Commission for Agricultural Costs and Prices (2020). *Price Policy for Sugarcane*.
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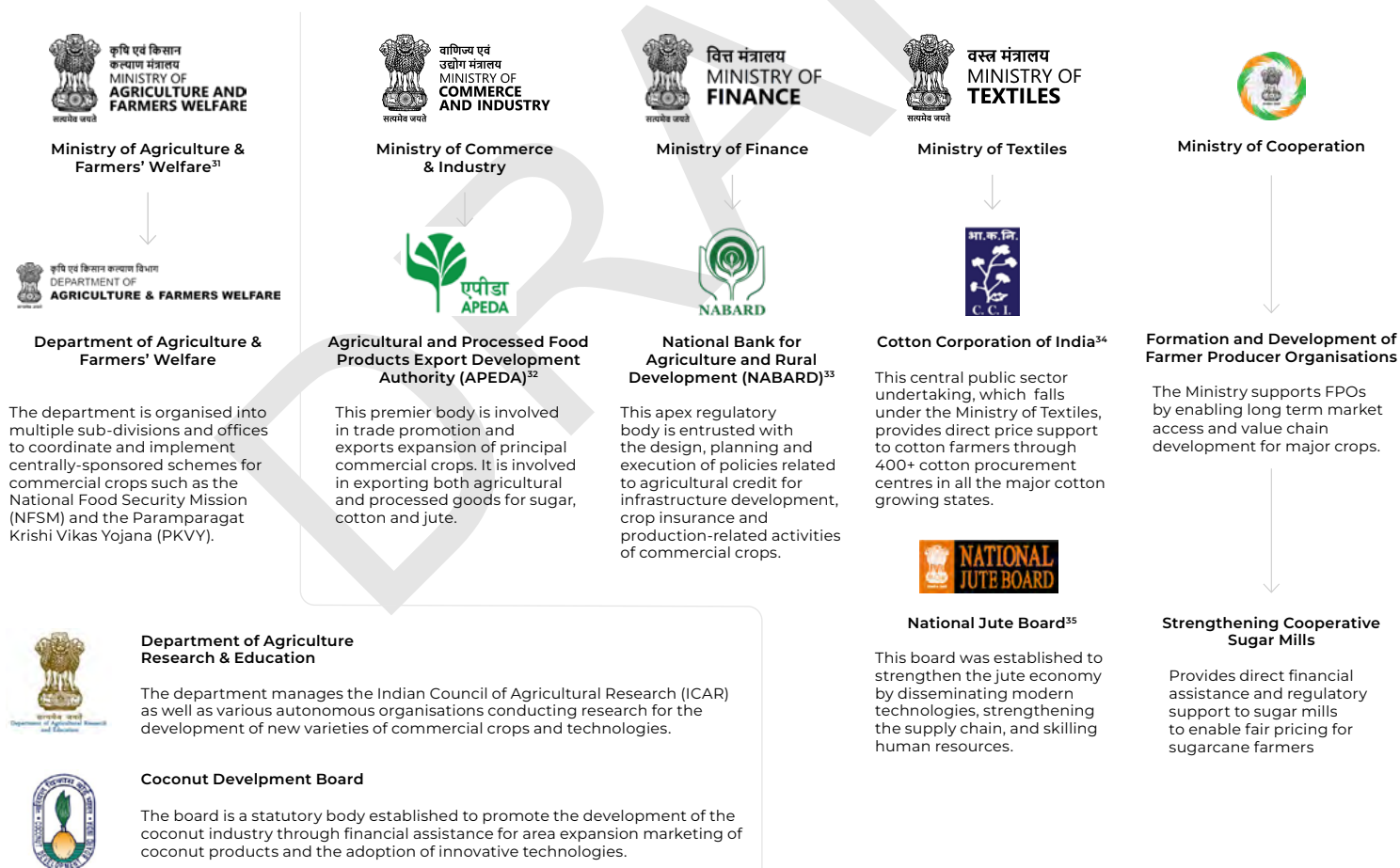
Water-Intensive Cultivation

- Both cotton and sugarcane are two of the most water intensive crops in India, requiring close to 29000 and 3000 litres respectively to produce 1kg.²⁸
- Production of commercial crops is incompatible with growing scarcity of water resources. More than 40% of agriculture in India is done in rainfed regions with low irrigation channels.²⁹ There is a greater need to accelerate adoption of methods like drip and sprinkler irrigation to make production more sustainable in future.

Rising Production Costs

- Both cotton and sugarcane are highly labor intensive which makes it challenging to maintain the required productivity against the growing scarcity of labour.
- Jute cultivation is affected by the lack of affordable quality, limited availability of good quality water in operations like retting, and high labour costs.²⁴
- Coconut cultivation is affected by rising labour costs, low product diversification and large harvest losses due to disease attacks and insufficient planting material. As a result, farmers are shifting away from coconut farming²⁷. The data from the Ministry of Agriculture and Farmers' Welfare reveals an 89% rise in the average cost of coconut production from 2017 to 2022.

Relevant Ministries and Departments



28 World Wide Fund for Nature. (2013). *Thirsty Crops*.

29 V Reddy et al, (2015). *Integrated Assessment of Scale Impacts of Watershed Intervention*.

30 Economic & Political Weekly. (2010). *Crop Insurance in India: Changes and Challenges*. Retrieved 5 October, 2023.

31 Ministry of Agriculture and Farmers' Welfare. Retrieved 23 July 2023.

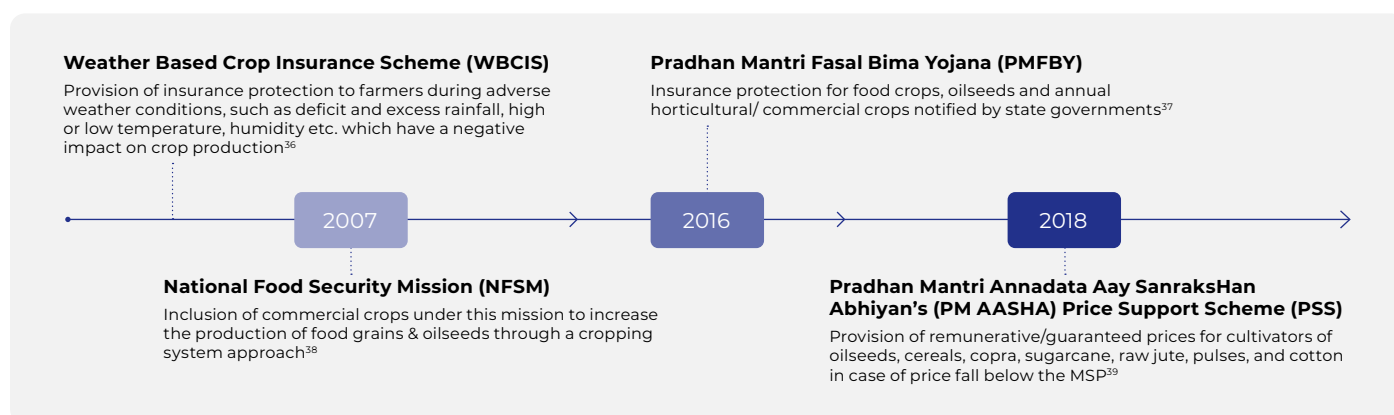
32 Agricultural and Processed Food Products Export Development Authority. (2012). *Cereals: The Grain Export Potential*.

33 Food and Nutritional Security in India. (2022). *Charting the way to a Robust Agri-food System*.

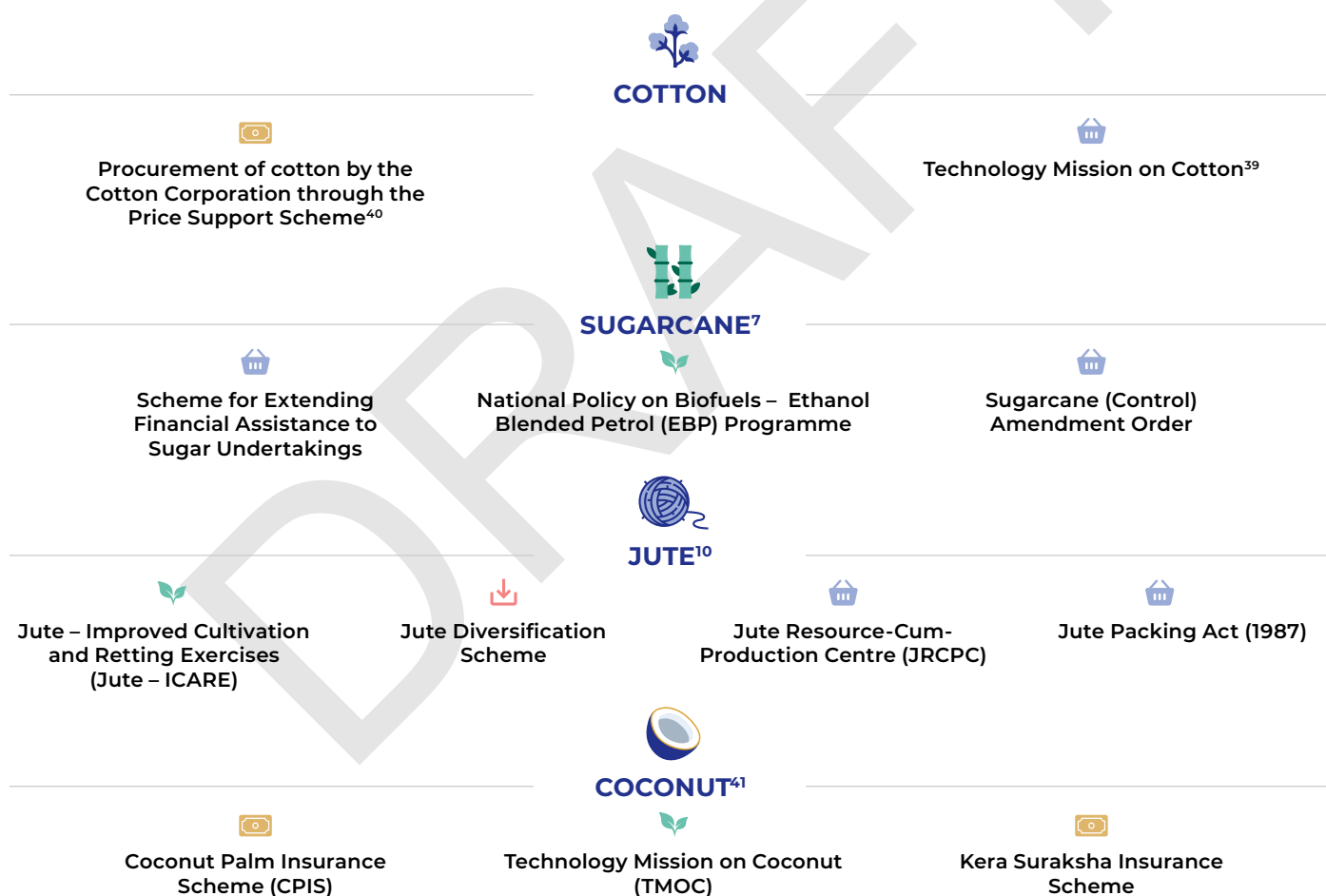
34 Cotton Corporation of India. Retrieved 18 August 2023.

35 Ministry of Textiles. Retrieved 18 August 2023.

Government policies such as the Weather Based Insurance Scheme (WBCIS), Pradhan Mantri Fasal Bima Yojana (PMFBY) and National Food Security Mission (NFSM) have been instrumental in shaping the Indian commercial crops sector.^{36 37 38 39}



As part of the overall policy landscape, crop-specific government interventions have been launched to aid capacity building, financial assistance, insurance, and technology missions for the development of specific commercial crops.



Challenges Addressed by Policy

Environmental Stressors for Production
 Market Challenges
 Financial Security
 Input-Intensive Cultivation

36 Ministry of Agriculture and Farmers' Welfare. (2014). *Restructured Weather Based Crop Insurance Scheme – Revised Operational Guidelines*.

37 National Insurance Company Ltd. Retrieved 18 August 2023.

38 National Food Security Mission Portal. Retrieved 18 August 2023.

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40 Ministry of Textiles. (2020). *National Textiles Policy, 2000*

41 Ministry of Textiles. (2023). *Demand No. 98*.

42 Coconut Development Board. *Brief guidelines of Coconut Development Board Schemes under implementation*. Retrieved 18 August 2023.



Sustainability and Product Diversification

Mainstream climate-resilient practices

- With growing consumer consciousness of eco-friendly products, there is a need to mainstream climate-smart practices across the commercial crop value chain. This includes more public-private partnerships, global exchange of practices, and direct financial support to agricultural enterprises engaged in last-mile adoption of precision farming technologies, organic and natural farming practices, and sustainable post-harvest management.
- Given India's groundwater extraction and water security challenges, cultivation practices for crops like sugarcane and cotton will need a greater thrust on drip irrigation, moisture stress tolerant varieties and intercropping.⁴³ States like Maharashtra and Andhra Pradesh have already tested pilot projects on cultivating sugarcane in drought-prone areas through drip irrigation.⁴⁴

Develop the value chain to effectively tap into emerging markets

- Both the sustainability and long-term economic viability of commercial crops such as coconut and jute depend on product diversification. High-quality processing and value addition through the modernisation of mills will enable a diverse range of products such as jelly and vinegar from coconut and woven and non-woven geotextiles from jute.⁴⁵



Inclusive and Competitive Value Chains

Promote gender-inclusive policies to maximise productivity⁴⁶

- Women play a vital role across all stages of cultivation including seed plantation, fertiliser application and weeding activities. According to the International Trade Center, women cotton farmers account for 70% of sowing labour and 90% of cotton-picking-labour in India. As social norms limit women's mobility and agency, Self Help Groups (SHGs) can play a critical role in providing women cotton farmers with easy access to knowledge and information, credit, extension services and other government benefits.

Addressing market inefficiencies in the sugarcane value chain

- High labour cost and input intensive cultivation associated with sugarcane cultivation can be mitigated by increased farm mechanisation. However, small and marginal farmers require greater financial and educational support to make use of modern farm implements. Going forward, it will be important to educate sugarcane harvesters on proven techniques of farm management and be provided with quality seeds, fertilizers & pesticides²⁶.
- Changing market preferences towards products like brown sugar and mineral sugar will have to be accounted for. Diversification of Indian sugar industry will be critical for long term sustainability and competitiveness²⁶.
- Modernisation of sugar mills will be paramount to ensure adequate sugar recovery and reduce post harvest losses²⁶.

Support small and marginal jute farmers with quality enhancement initiatives

- As low output quality puts a disproportionate number of small jute farmers at a disadvantage, there is a need to prioritise capacity building and technologies like microbial formulation to significantly improve post-harvest practices like retting. Through the Technology Mission on Jute, multiple solutions like efficient microbial consortia and mechanical extraction processes have been successfully demonstrated in West Bengal.⁴⁷

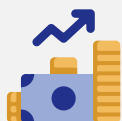
43 S K Shukla et al. (2020). Low cost Technologies in Sugarcane Agriculture.

44 F A Zainol et al. (2023). Coconut Value Chain Analysis.

45 S Shahinur et al. (2022). Current Development and Future Perspective on Natural Jute Fibers and Their Biocomposites.

46 The Sustainable Trade Initiative & Sattva Consulting. (2019). *Business case for gender mainstreaming in cotton in Maharashtra*.

47 B Majumdar et al. (2017). Comparative Study of Conventional and Improved Retting of Jute with Microbial Formulation.



Financial Security of Farmers

Collectivise farmers to enable formal credit linkages

- Farmers do not produce copra/oil for sale for a variety of reasons, the primary one being the less marketable surplus given the small and marginal size of their farms.⁴⁷ In the case of sugarcane harvesters, rising production costs increase the risk of debt trap.⁴⁸ To protect farmer incomes and improve farm productivity, crop insurance and working capital loans can be delivered through Farmer Producer Organisations (FPOs) and Companies (FPCs).
- A NABARD study found that numerous FPOs in states like Rajasthan, Jharkhand, Chhattisgarh and Tamil Nadu have succeeded in leveraging various local government schemes to establish successful credit linkages with banks and other financial institutions.⁴⁹



Technological Development

Improve harvest timing and cultivation operations

- Optimal harvesting timing plays a vital role in ensuring crop productivity, making it necessary to support farmers with technology that can help predict appropriate harvesting times to maximise outputs.⁴⁷
- The Sub Mission on Agricultural Mechanisation scheme provides financial assistance to set up high-tech machinery hubs for commercial crops like sugarcane and cotton. Private sector capabilities can be strongly leveraged through innovative business models to introduce a diverse range of equipment for supporting the entire life cycle of crop production. In Punjab, an attempt was made to train rural youth in machinery usage for various post-harvest operations and promote local entrepreneurship.⁵⁰

Reduce post-harvest crop loss

- Sugarcane tends to lose its sugar content within a few days after harvest which poses serious financial implications for milling operations. This problem is likely to increase with prolonged droughts and rising temperatures in top producing states like Uttar Pradesh, Karnataka and Maharashtra.⁵¹ With a surge in development of precision farming methods, modern technological tools can play a vital role in curbing the issue by enabling robust monitoring mechanisms.
- Funds have been earmarked in the Agriculture Infrastructure Fund to support cooperatives and entrepreneurs with developing post-harvest infrastructure. Due to the consistent growth of startups in India, agricultural technology like drones and artificial intelligence can play a crucial role in improving the transportation and inventory management of sensitive crops like sugarcane.⁵² For instance, in Brazil, sugarcane harvest dates have been determined based on remote sensing and satellite imagery.⁵³

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49 Department of Economic Analysis and Research & National Bank for Agriculture and Rural Development. (2019-21). Case Studies of FPOs in India (2019-2021).

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